

Cagrilintide Available for Research Use Only

(0.3 – 4.5 mg once weekly, dose-escalated)

Cagrilintide is a long-acting amylin analogue developed for potent appetite suppression and weight loss; Phase 2 and Phase 3 clinical data show substantial weight reduction versus placebo and active comparators, and extensive preclinical work maps its brainstem/hypothalamic satiety mechanisms.

 Probably safe (substantial human evidence)

 Effective (human trials show effectiveness, combined with good understanding)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Cagrilintide has substantial human clinical evidence as both monotherapy and in combination with semaglutide. Phase 2 monotherapy studies demonstrated **dose-dependent weight loss** with once-weekly Cagrilintide, while subsequent Phase 3 CagriSema trials confirmed **substantial weight loss when cagrilintide 2.4 mg was combined with semaglutide 2.4 mg once weekly.**

In REDEFINE 1, adults with overweight or obesity without diabetes receiving CagriSema 2.4/2.4 mg once weekly had an estimated mean body-weight reduction of approximately 20% under the treatment-policy estimand at 68 weeks. Cagrilintide 2.4 mg and semaglutide 2.4 mg monotherapy arms were also included, confirming **greater weight loss with the combination than with either component alone.**

In the 2026 REDEFINE 4 head-to-head trial, CagriSema did not demonstrate noninferiority to tirzepatide for weight reduction (23.0% vs 25.5% under the trial-product estimand at 84 weeks).

Longer-term cardiovascular-outcome and rare-event safety characterization remains incomplete despite the availability of large Phase 3 safety datasets.

Dose

Cagrilintide is a **long-acting amylin analogue designed for once-weekly subcutaneous administration.** Its terminal elimination half-life is approximately **7–8 days**, supporting weekly rather than daily dosing.

Doses Studied as Monotherapy

The Phase 2 dose-ranging obesity trial evaluated:

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- **0.3 mg once weekly**
- **0.6 mg once weekly**
- **1.2 mg once weekly**
- **2.4 mg once weekly**
- **4.5 mg once weekly**

Weight loss was dose-dependent, with progressively greater mean weight reduction across the dose range. The 4.5 mg dose produced the greatest weight reduction in the Phase 2 monotherapy study, but GI adverse events were common and generally dose-related.

The clinical studies used **gradual dose escalation rather than beginning immediately at the target dose**, primarily to improve gastrointestinal tolerability.

Dose Used with Semaglutide

The principal CagriSema regimen studied in Phase 3 is:

- **Cagrilintide 2.4 mg + semaglutide 2.4 mg SC once weekly**

The components were dose-escalated toward the target dose rather than initiated at full dose.

The 2.4 mg cagrilintide dose should therefore be regarded as the best-characterized current dose for combination therapy with semaglutide, while cagrilintide monotherapy has been studied at doses as high as 4.5 mg weekly.

Dose Escalation

Slow titration is important because gastrointestinal intolerance is dose-related. Nausea, vomiting, constipation, reduced appetite, and early satiety are the principal tolerability limitations.

The available trials support the general principle:

low starting exposure → gradual escalation → individually tolerated maintenance exposure

rather than rapid escalation to the maximum studied dose.

Specific titration schedules should be identified by trial when discussed because monotherapy and CagriSema studies have not all used identical escalation schemes.

Frequency

Once weekly is the appropriate research-based dosing frequency.

Because cagrilintide has an approximately 7–8-day terminal half-life, there is no established pharmacologic rationale for routine daily administration. Likewise, splitting a weekly dose into multiple injections per week has not been established as superior to once-weekly administration in the major clinical trials.

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Dose Summary

Use	Cagrilintide exposure studied
Phase 2 monotherapy	0.3–4.5 mg once weekly
Higher-dose Phase 2 monotherapy	4.5 mg once weekly
CagriSema Phase 3	2.4 mg once weekly
Semaglutide component of CagriSema	2.4 mg once weekly
Administration	Subcutaneous
Terminal half-life	Approximately 7–8 days
Routine frequency studied	Once weekly

These are studied doses, not an established therapeutic dose range; 4.5 mg was the highest dose in the Phase 2 monotherapy dose-ranging trial.

Benefits

- **Dose-dependent weight loss** — demonstrated in human monotherapy trials.
- **Marked appetite suppression and increased satiety** — supported by human and mechanistic data.
- **Reduced meal size** — consistent with amylin-receptor pharmacology and experimental evidence.
- **Greater weight loss when combined with semaglutide** — demonstrated in Phase 2 and Phase 3 trials.
- **Complementary amylin/GLP-1 appetite regulation** — strong mechanistic rationale.
- **Improved glycemic parameters with CagriSema** — demonstrated particularly in subjects with impaired glucose regulation or type 2 diabetes.

Indications

- **Obesity or overweight with comorbidities** — *Human clinical trials.*
- **Combination with GLP-1 receptor agonism for greater weight reduction** — *supported directly for cagrilintide + semaglutide.*
- **Research use in metabolic physiology studies** — *Preclinical and translational research.*

Contraindications

- **Pregnancy and breastfeeding** — *Standard for weight-loss agents; theoretical fetal risk.*
- **Uncontrolled eating disorders** — *Clinical caution; may worsen disordered intake.*
- **Severe gastroparesis** — *Because of further gastric emptying delay; mechanistic concern.*
- **Unstable cardiovascular disease** — *Limited trial data; exercise clinical caution.*

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Side effects

- **Nausea and transient vomiting** — *Human clinical data; dose-related.*
- **Early satiety and reduced caloric intake** — *Expected effect; can cause dizziness or weakness in some.*
- **Possible heart-rate increase/palpitations** - *monitor as a potential cardiovascular effect.*
- **Possible hypoglycemia when combined with insulin or insulin secretagogues** — *Mechanistic risk; clinical vigilance required.*

Biological mechanism

Cagrilintide is a long-acting amylin analogue that activates calcitonin receptor/receptor activity-modifying protein complexes forming amylin receptors, including AMY1, AMY2, and AMY3 receptors.

Its weight-loss effects appear to arise predominantly through central regulation of appetite, satiation, and meal termination. Important neural targets include the dorsal vagal complex and associated brainstem/hypothalamic circuits.

Experimental work indicates that AMY1 and AMY3 receptor signaling contributes importantly to cagrilintide-induced weight loss.

Cagrilintide and GLP-1 receptor agonists act through complementary neural and metabolic pathways. This provides a mechanistic basis for the greater weight reduction observed when cagrilintide is combined with semaglutide.

The peptide has an elimination half-life of approximately 7–8 days, permitting sustained receptor exposure with once-weekly administration.

References

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